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What is Testing?

Testing in Software Development is writing code to verify weather some functionality works as expected

Testing is especially useful for larger applications where we wish to verify the application is working even after certain changes were made to some parts of the application code

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Testing Benefits

Testing provides several benefits

- Quality
- Stability
- Fast Feedback
- Better Design

Improves the quality of the code written for the applications

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Testing Benefits

Testing provides several benefits

- Quality
- Stability
- Fast Feedback
- Better Design

It allows code base to evolve with stability and ensure working functionality

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Testing Benefits

Testing provides several benefits

- Quality
- Stability
- Fast Feedback
- Better Design

It allows to get immediate feedback on a broken piece of important functionality and it's source

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Testing Benefits

Testing provides several benefits

- Quality
- Stability
- Fast Feedback
- Better Design

It forces us to modularize our source code and create clear separation of concerns for better design

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Types of testing

There are several types of testing in software development. Some of these tests are performed by testers and others by developers

When developing applications developers usually write three types of teste, these are : unit tests, integration tests and acceptance tests

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Types of testing

There are three types of tests that developers write:

- Unit
- Integration
- Functional

Unit testing is used to test individual components in isolation

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Types of testing

There are three types of tests that developers write:

- Unit
- Integration
- Functional

Integration Testing is used to test weather our components are working correctly together

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Types of testing

There are three types of tests that developers write:

- Unit
- Integration
- Functional

Functional tests are used to automate a use case and check if it's working correctly

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Unit Test

Unit testing is used to test each individual unit of our application is isolation

These units can be a small utility function an object etc. The point is these units should be as small as possible and isolated

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Unit Testing Frameworks

Unit Testing can be very tiresome to write completely from scratch, to overcome this issue there are many testing frameworks

Unit Testing Frameworks allow you to write code and make assertions about the data that is returned or how the parameters are manipulated etc.

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Unit Testing Frameworks

There are several testing frameworks available for JavaScript

- Jest
- QUnit
- Mocha
- Ava
- Jasmine

Jest was developed by Facebook and is one of the most popular unit testing frameworks especially in react

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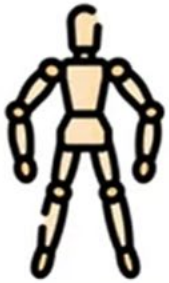
Test Doubles

When doing unit testing we need to isolate the modules, but that can be difficult as some modules depend on other modules

To overcome this problem we use fake implementations of those modules, e.g. in-memory database, fake http server etc.



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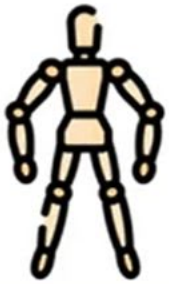

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Test Doubles

There are several types of Test Doubles

- Mocks
- Stubs
- Spies
- Fakes
- Dummies

Mocks are objects we can use to verify method calls that are expected.

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Test Doubles

There are several types of Test Doubles

- Mocks
- Stubs
- Spies
- Fakes
- Dummies

Stub is an object that holds predefined data and returns it whenever it's called during tests

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Test Doubles

There are several types of Test Doubles

- Mocks
- Stubs
- Spies
- Fakes
- Dummies

Fakes are objects that have working implementations, but not same as production one

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Test Doubles

There are several types of Test Doubles

- Mocks
- Stubs
- Spies
- Fakes
- Dummies

Spies are stubs that also record some information based on how they were called

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Test Doubles

There are several types of Test Doubles

- Mocks
- Stubs
- Spies
- Fakes
- Dummies

Dummy objects are passed around but never actually used. Usually they are just used to fill parameter lists

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Integration Testing

Integration Tests are quite different from unit tests as in integration tests we check weather a module interacts properly with other modules in our system

Integration Tests are done on modules that are interacting with external modules or external systems like, database packages, cache systems, blob storage etc.

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Integration Testing

IntelliPaat

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Integration Testing Advantages

Integration Testing has several advantages

Faster development

Stable Integration

Easy Error Detection

It accelerates development as we get immediate feedback if the current code integrates correctly

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Integration Testing Advantages

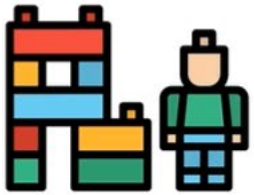
Integration Testing has several advantages

Faster development

Stable Integration

Easy Error Detection

It allows developers to integrate a new module easily



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Integration Testing Advantages

Integration Testing has several advantages

Faster development

Stable Integration

Easy Error Detection

If changes in a module breaks some functionality that relates to intefration we would know about it immediately

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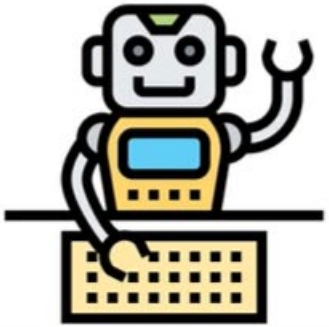
Functional Testing

Functional Testing is used to test out an entire user story by automating a system to interact with your application

Functional Testing helps us check if a particular feature is working from a users perspective



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Functional Testing Framework

Application interaction can be a little difficult to automate and teste which is why there are several Functional Testing Frameworks out there

These frameworks allow us to write code to automate an entire user story

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Functional Testing Framework

Testing provides several benefits

- Cypress
- Puppeteer
- Protractor
- Selenium

Cypress is a new end to end testing framework that allows developers to compare snapshots of each step in user story

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Functional Testing Framework

Testing provides several benefits

- Cypress
- Puppeteer
- Protractor
- Selenium

Puppeteer is a Node library which provides a high-level API to control Chrome or Chromium over the DevTools Protocol

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Functional Testing Framework

Testing provides several benefits

- Cypress
- Puppeteer
- Protractor
- Selenium

Protractor is an end-to-end test framework for Angular and AngularJS applications

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Functional Testing Framework

Testing provides several benefits

- Cypress
- Puppeteer
- Protractor
- Selenium

Selenium is a browser automation library. Most often used for testing web-applications

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